

★ WEEK 3 · PART 3 of 3

Selection, Diagnostics & Forecasting

Validating a Model — and Knowing Its Limits

AIC/BIC · residual diagnostics · forecasting · the humbling truth about returns

A fitted model is not a finished model. Part 3 completes the loop: choose among candidates, prove the residuals are clean, forecast honestly — and confront why ARMA finds so little in raw returns.

3

PART 3 · VALIDATION

From a Fit to a Forecast

Model selection with AIC and BIC, diagnostic checking of the residuals, forecasting with honest intervals, and a clear-eyed look at what ARMA can and cannot predict.

Closing the Box-Jenkins Loop

Part 2 fit models; Part 3 validates and uses them

1

We can identify and estimate an ARMA

The ACF/PACF propose a candidate; the software fits it. But a fit is only a hypothesis until it survives scrutiny.

2

Now we select, check, and forecast

Compare candidates with AIC/BIC, confirm the residuals are white noise, and only then produce forecasts with honest uncertainty.

3

And we stay humble

ARMA is powerful where linear memory exists — and nearly useless on raw returns. Knowing the difference is the mark of an expert.

Part 3 in one line: prove the model is clean, forecast honestly, and know its limits.

AIC and BIC, Revisited

Two penalties, two philosophies

Both criteria reward fit and punish complexity — they differ only in how harshly. AIC aims for the best forecaster; BIC aims for the true, simplest model.

AIC — the forecaster's choice

- A lighter penalty ($2k$) tolerates a few extra terms.
- It targets out-of-sample predictive accuracy.
- It can occasionally keep a parameter that only fits noise.

BIC — the truth-seeker's choice

- A heavier penalty ($k \cdot \ln T$) that grows with sample size.
- It is consistent: it finds the true order as data grows.
- It leans toward simpler, more parsimonious models.

AIC vs BIC: Which and When

A quick decision guide

Criterion	Penalty on k	Tends to select	Prefer it when
AIC	$2k$ (lighter)	slightly larger models	your goal is forecasting
BIC	$k \cdot \ln T$ (heavier)	simpler models	your goal is the 'true' order

When they disagree, default to the simpler (BIC) model unless forecasting clearly needs the extra terms.

Are the Residuals White Noise?

The final gate before trusting a model

A model has captured the structure only if what it leaves behind — the residuals — is indistinguishable from white noise. Any leftover pattern means the job is not done.

What 'clean' means

- No autocorrelation left in the residuals at any lag.
- Roughly constant variance and no obvious outliers.
- Nothing the model could still have exploited.

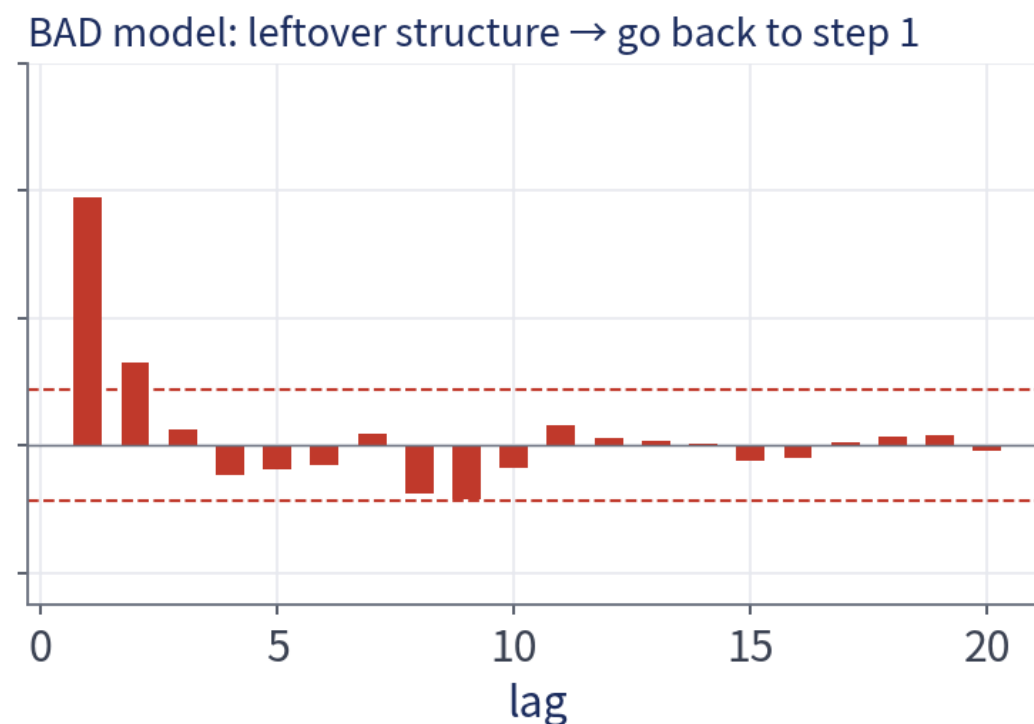
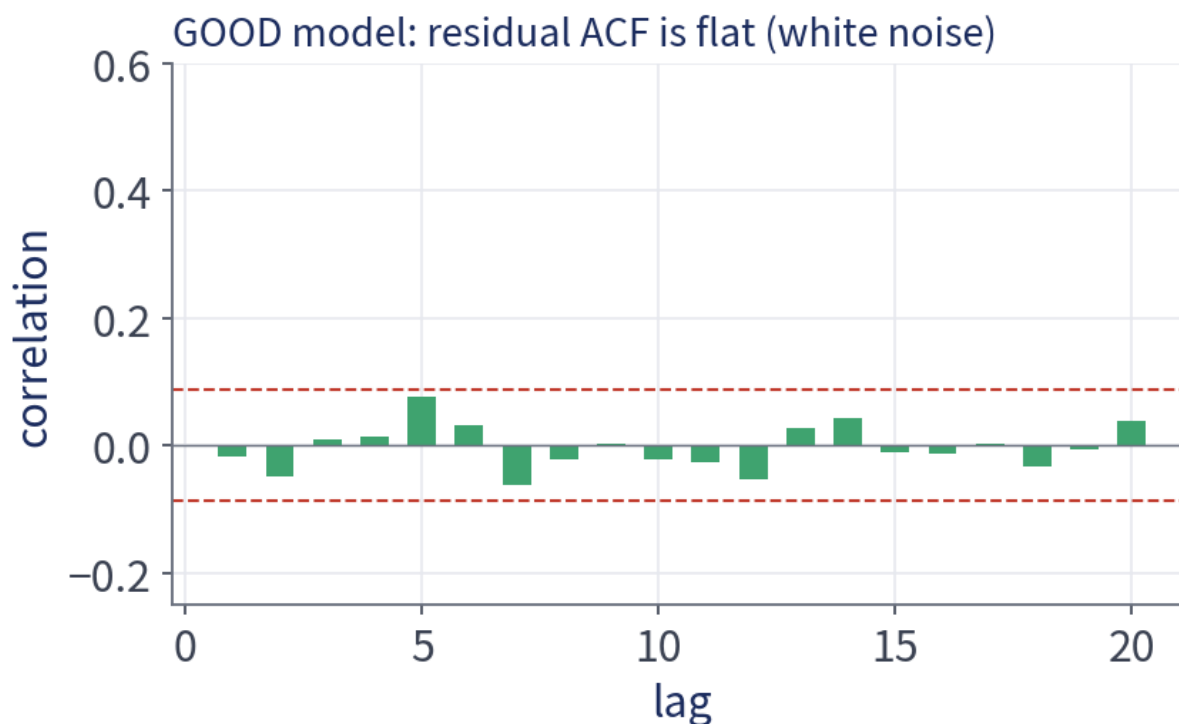
Why it is the exit test

- If structure remains, the forecast is leaving signal on the table.
- A leftover spike tells you exactly where to add order.
- Only white-noise residuals let you leave the loop.

Reading the Residual ACF

A good model leaves a flat residual ACF (left). A leftover spike (right) means the model missed structure — go back and raise p or q .

The final gate: are the residuals indistinguishable from white noise?



The Ljung-Box Test

A formal check for leftover autocorrelation

$$Q = T(T + 2) \sum_{k=1}^h \frac{\hat{\rho}_k^2}{T-k} \sim \chi_{h-m}^2 \quad (\text{Ljung-Box})$$

What it does

- Combines the first h residual autocorrelations into one statistic.
- Under a clean model it follows a chi-squared distribution.
- A small p-value means the residuals still carry structure.

How to use it

- Run it on the residuals after fitting.
- A LARGE p-value is good — it means 'looks like white noise'.
- Pair it with the residual ACF plot; never rely on one alone.

Other Diagnostic Checks

White noise is necessary, not sufficient

Also inspect

- Normality of residuals: a Q-Q plot or a Jarque-Bera test.
- Constant variance: watch for volatility clustering in residuals.
- Stability: do the coefficients hold across sub-samples?

What they reveal

- Fat-tailed residuals warn that intervals may be too narrow.
- Clustered residual variance points to a GARCH model (later).
- Unstable coefficients hint at a structural break.

Forecasting With ARMA

Turn the fitted model forward in time

To forecast, feed the model's own predictions back into it, step by step. Each step uses known values and past forecasts — and expected future shocks are simply zero.

The mechanism

- One step ahead uses the last observed values directly.
- Further steps substitute earlier forecasts for unknown values.
- Unknown future shocks contribute their expected value: zero.

The consequence

- The influence of what you know fades with the horizon.
- So long-horizon forecasts converge to the series' mean.
- The next slides make this concrete.

Forecasts Decay to the Mean

The farther out you look, the less the past helps

$$\hat{X}_{T+h} = \mu + \phi^h (X_T - \mu) \xrightarrow{h \rightarrow \infty} \mu \quad (\text{AR}(1) \text{ forecast})$$

What the formula says

- The forecast is the mean plus a shrinking piece of today's deviation.
- That piece shrinks like ϕ^h as the horizon h grows.
- Beyond a few steps, the best guess is essentially the mean.

Why it makes sense

- A stationary series has no lasting trend to extrapolate.
- Its memory of any shock fades — so must the forecast's edge.
- This is mean reversion, seen through the forecast.

Forecasts Need Honest Intervals

A point forecast alone is dangerously incomplete

$$\hat{X}_{T+h} \pm z_{0.975} \sigma_h, \quad \sigma_h \text{ grows with the horizon } h$$

Why intervals matter

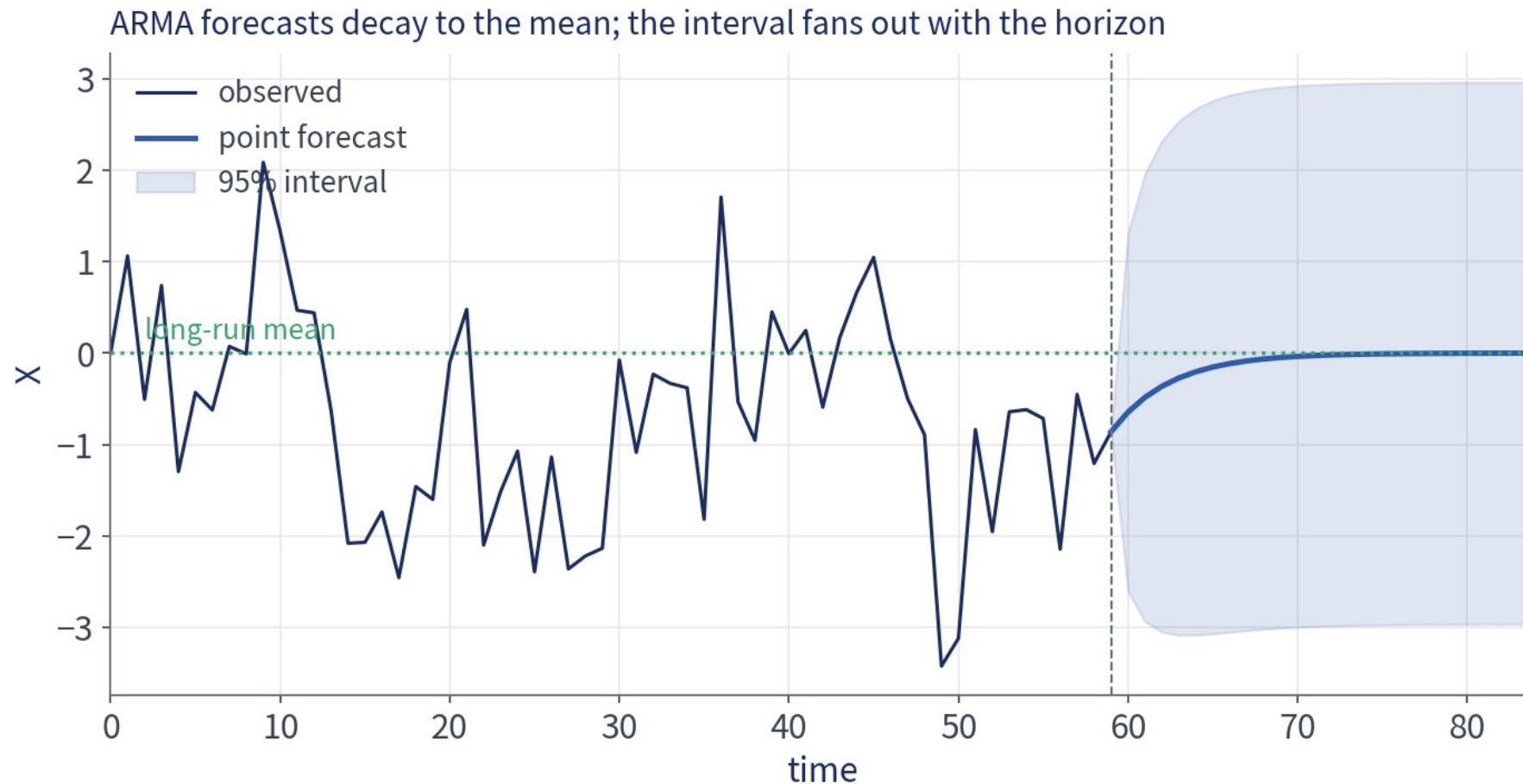
- A single number hides how uncertain the forecast really is.
- The interval σ_h widens as the horizon grows.
- It reflects the accumulation of future unknown shocks.

How to read them

- Near term: tight bands — recent data pins the value down.
- Far term: wide bands centered on the mean.
- Report the interval, not just the point, every time.

The Forecast Fan

The point forecast (blue) decays to the long-run mean while the 95% interval fans out — a compact picture of everything ARMA forecasting does.



How Far Ahead Is Useful?

Knowing when a forecast stops saying anything

An ARMA forecast is informative only while its point estimate still differs meaningfully from the mean. Past that horizon, you are just quoting the average.

The practical limit

- Persistence sets the useful horizon: higher ϕ reaches farther.
- Once ϕ^h is tiny, the forecast is indistinguishable from the mean.
- The interval, meanwhile, has widened to its long-run width.

The honest takeaway

- Short horizons: genuine, usable information.
- Long horizons: the mean, dressed up with wide bands.
- Match your forecast horizon to the series' memory.

In-Sample vs Out-of-Sample

The only test that really counts

A model can fit the past beautifully and still forecast the future badly. The verdict that matters is out-of-sample: performance on data the model never saw while fitting.

The trap of in-sample fit

- In-sample error always falls as you add parameters.
- That improvement can be pure overfitting.
- It flatters complex models that will not generalize.

The out-of-sample discipline

- Hold out the most recent data; fit on the rest.
- Forecast the held-out period and measure the error.
- Or roll the window forward for a rigorous back-test.

Forecast Error Metrics

How to score a forecast

Metric	What it measures	Character
RMSE	root mean squared error	penalizes large misses heavily
MAE	mean absolute error	average miss; robust to outliers
MAPE	mean absolute percentage error	scale-free, but unstable near zero

Always score against a naive benchmark (e.g. 'tomorrow = today'), never against zero error.

The Benchmark You Must Beat

A forecast is only good if it beats the obvious

For many financial series the naive forecast — 'tomorrow equals today' — is remarkably hard to beat. If your ARMA cannot beat it out of sample, it adds nothing.

The naive benchmarks

- Random walk: next value equals the last observed value.
- Historical mean: next value equals the long-run average.
- Seasonal naive: next value equals the same period last cycle.

The honest comparison

- Report your model's error AND the benchmark's error.
- Beating the benchmark is the real evidence of value.
- Failing to beat it is a useful, humbling result.

A Humbling Truth About Returns

ARMA describes structure — but returns have little

Apply the entire Box-Jenkins machine to raw asset returns and you find almost nothing: their autocorrelation is essentially zero. The direction of returns is not linearly predictable.

Why this happens

- Efficient markets arbitrage away simple linear patterns (Week 1).
- So the ACF of returns sits inside the confidence bands.
- An ARMA on returns fits essentially white noise.

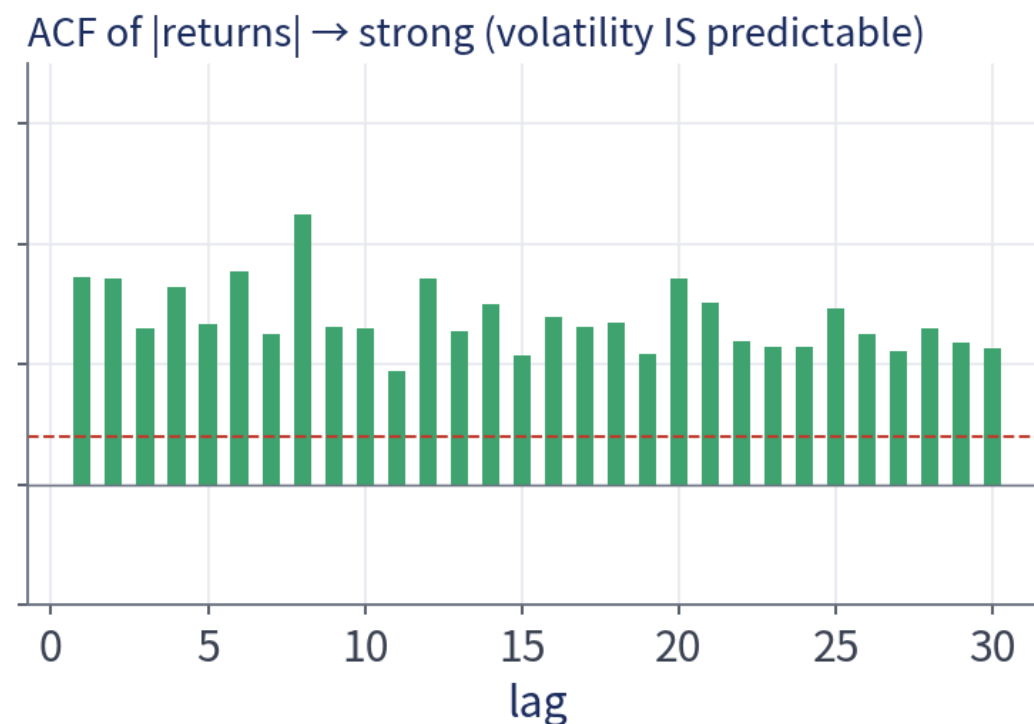
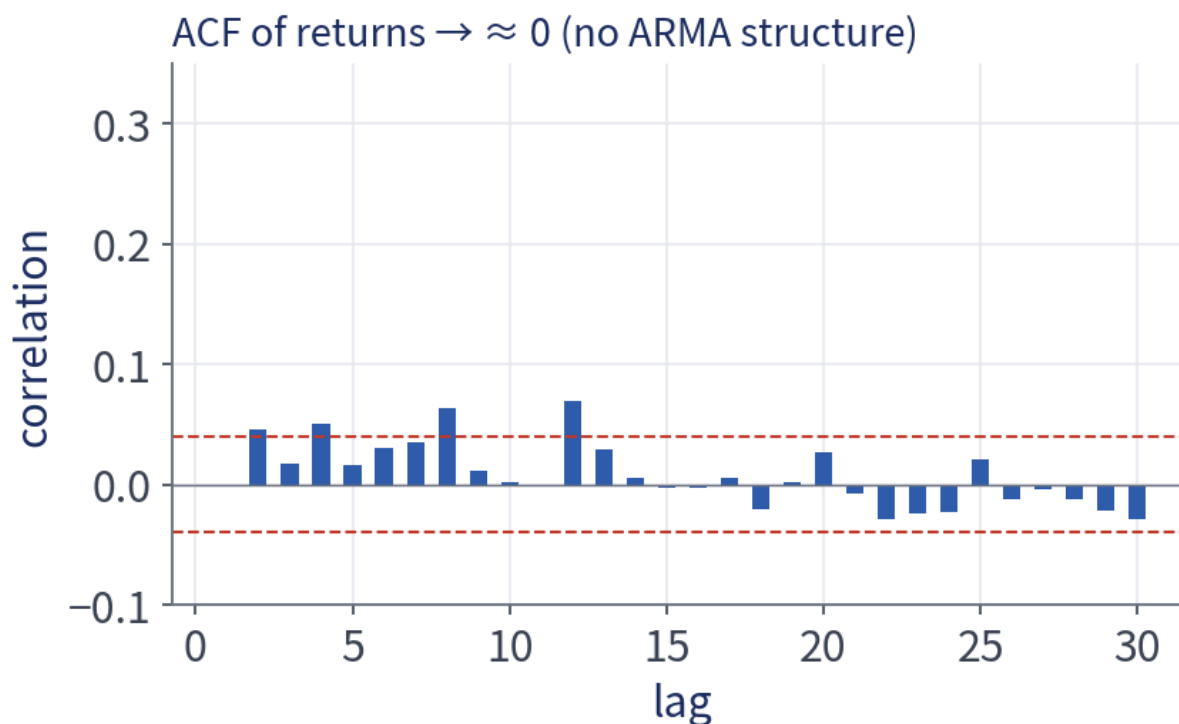
Why it is not defeat

- It tells you where NOT to spend effort — direction forecasting.
- It redirects you to what IS predictable.
- That redirection is the most valuable lesson of the week.

Returns Resist ARMA — But Volatility Does Not

Left: the ACF of returns is essentially zero, so ARMA finds nothing. Right: the ACF of $|\text{returns}|$ is strongly positive — the predictable structure lives in volatility.

The humbling truth: ARMA finds little in returns — the structure lives in volatility



So What IS Predictable? Volatility

The door ARMA opens onto later weeks

$$\underbrace{\rho_r(k) \approx 0}_{\text{returns: no ARMA}} \quad \underbrace{\rho_{|r|}(k) > 0}_{\text{volatility: predictable}}$$

The split, one more time

- The level of returns: no linear memory — ARMA is powerless.
- The magnitude of returns: strong, persistent memory.
- Volatility clusters, and clustering is exactly what we can model.

Where it leads

- Model the variance, not the mean: ARCH and GARCH.
- The same ACF logic, applied to squared or absolute returns.
- Week 3's tools carry straight over to volatility.

Where ARMA Helps — and Where It Does Not

ARMA earns its keep on series with genuine linear memory; it finds little in raw returns, but points the way to volatility models.

Where ARMA Helps - and Where It Does Not

A model is only as good as the structure that is actually there to find.

ARMA earns its keep here

Series with genuine linear memory:

- macro levels & growth rates
- interest rates and spreads
- inflation, demand, inventories
- cointegrated spreads (Week 2)
- seasonal / trending data (Week 4)

These have autocorrelation the ACF / PACF can see - so ARMA can exploit it.

The humbling truth: raw returns

Asset returns are almost uncorrelated:

- ACF of returns is essentially 0
- so ARMA finds little to model
- direction is not linearly predictable

But not all is lost:

The ACF of $|returns|$ is strongly positive - volatility IS predictable.

That is the door to GARCH (later weeks).

The Complete Modeling Checklist

Weeks 2 and 3 in one routine

1 · Stationarize

Week 2

Difference or detrend;
confirm with ADF/KPSS.

2 · Identify

ACF/PACF

Propose candidate (p, q)
from the fingerprints.

3 · Estimate

fit + rank

Fit candidates;
compare by AIC and BIC.

4 · Diagnose

residuals

Check the residual ACF
and Ljung-Box.

5 · Forecast

honestly

Predict with intervals;
beat a naive benchmark.

One disciplined pass from raw data to a validated, honestly-evaluated forecast.

Forecast, Evaluate, and Confront Returns

Direct Claude Code from a fitted model to an honest verdict

GOAL Forecast a fitted ARMA with intervals, back-test it against a naive benchmark, then test the humbling truth on real returns.

[INPUT] Prompt · Claude Code

Write and run `forecast_lab.py` that:

1. fits your chosen ARMA (from Lab 2) and produces an $h=20$ forecast with 95% bands;
2. back-tests it out-of-sample vs a random-walk benchmark, reporting RMSE for both;
3. then fits an ARMA to real daily RETURNS and plots the ACF of returns vs $|returns|$.

Tell me: does the ARMA beat the naive benchmark, and does it find any structure in the returns themselves?

How to work with it

- The forecast should fan out and revert to the mean.
- Beating the random walk is hard — note if it fails.
- The returns' ACF should look flat; $|returns|$ should not.
- This closes the Week-3 arc and opens volatility.

Deliverable: `forecast_lab.py` + a forecast fan and the returns-vs- $|returns|$ ACF.

Three Questions Before You Move On

If you can answer these, Part 3 has landed

Q1

A Ljung-Box test on your residuals gives $p = 0.02$. Good or bad news?

Bad — structure left

Q2

You forecast an AR(1) 30 steps ahead. What does the point forecast approach?

The mean

Q3

Your ARMA on returns has a near-flat ACF. What does that tell you?

No linear structure

All three follow from diagnostics, mean reversion, and the humbling truth.

Three Pitfalls to Avoid

Where forecasting goes wrong

Common mistakes

- "A great in-sample fit means great forecasts."
- "Report the point forecast; the interval is optional."
- "ARMA should be able to predict returns."

The reality

- Only out-of-sample performance counts; in-sample flatters complexity.
- The interval is the forecast — a point alone hides the uncertainty.
- Raw returns are ~unpredictable; model volatility instead.

Week 3 in Three Ideas

Pulling the whole week together

1

AR and MA are two engines; ARMA fuses them

Part 1: AR remembers levels, MA remembers shocks, and Wold's theorem plus $AR \leftrightarrow MA$ duality tie them together. The ACF/PACF are the fingerprints.

2

Box-Jenkins turns modeling into a disciplined loop

Part 2: identify with the cheat sheet, estimate, select by AIC/BIC, and diagnose — repeating until the residuals are white noise.

3

Forecast honestly, and know ARMA's limits

Part 3: forecasts revert to the mean with widening intervals; ARMA finds little in raw returns, pointing us toward volatility models next.

Next — Week 4: ARIMA and Seasonality, extending ARMA to trending and seasonal data.